

GS 21/L

PHYSICAL AND STRUCTURAL GEOLOGY



Chapter 9: Groundwater



GROUNDWATER

- It is the largest reservoir of fresh water available to human. It is the water in the zone of saturation.

Importance of groundwater

1. Invaluable source of fresh water for the following:

- Industry
- Agriculture
- Human well-being

2. Important erosional agent- responsible for formation of sinkholes and doline.

3. Acts as storage of meteoric water during precipitation.

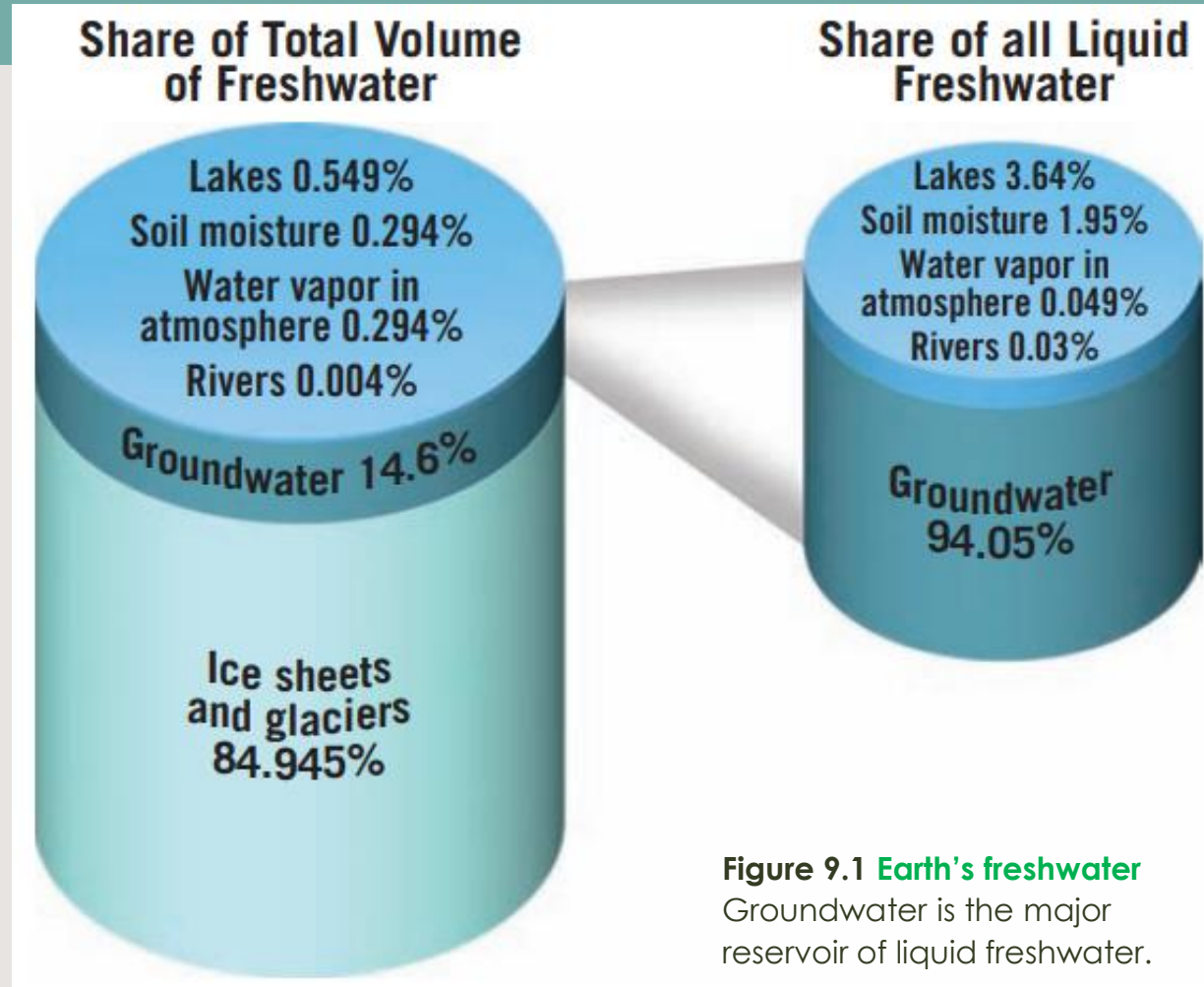


Figure 9.1 Earth's freshwater
Groundwater is the major reservoir of liquid freshwater.

Groundwater

Distribution of Ground Water

Different zones:

1. **Zone of soil moisture**- uppermost layer of soil with moisture from meteoric water. Near surface zone where some of the water that soaks in does not travel far because it is held by molecular attraction as a surface film on soil particles.
2. **Vadose zone (Unsaturated zone)** – the area above the water table that includes the capillary fringe and the zone of soil moisture. The pore spaces in this zone contain both air and water.
3. **Phreatic zone (saturated zone)** – a zone where water that is not held as soil moisture percolates downward reaches, where all the open spaces in sediment and rock are completely filled with water.

Groundwater

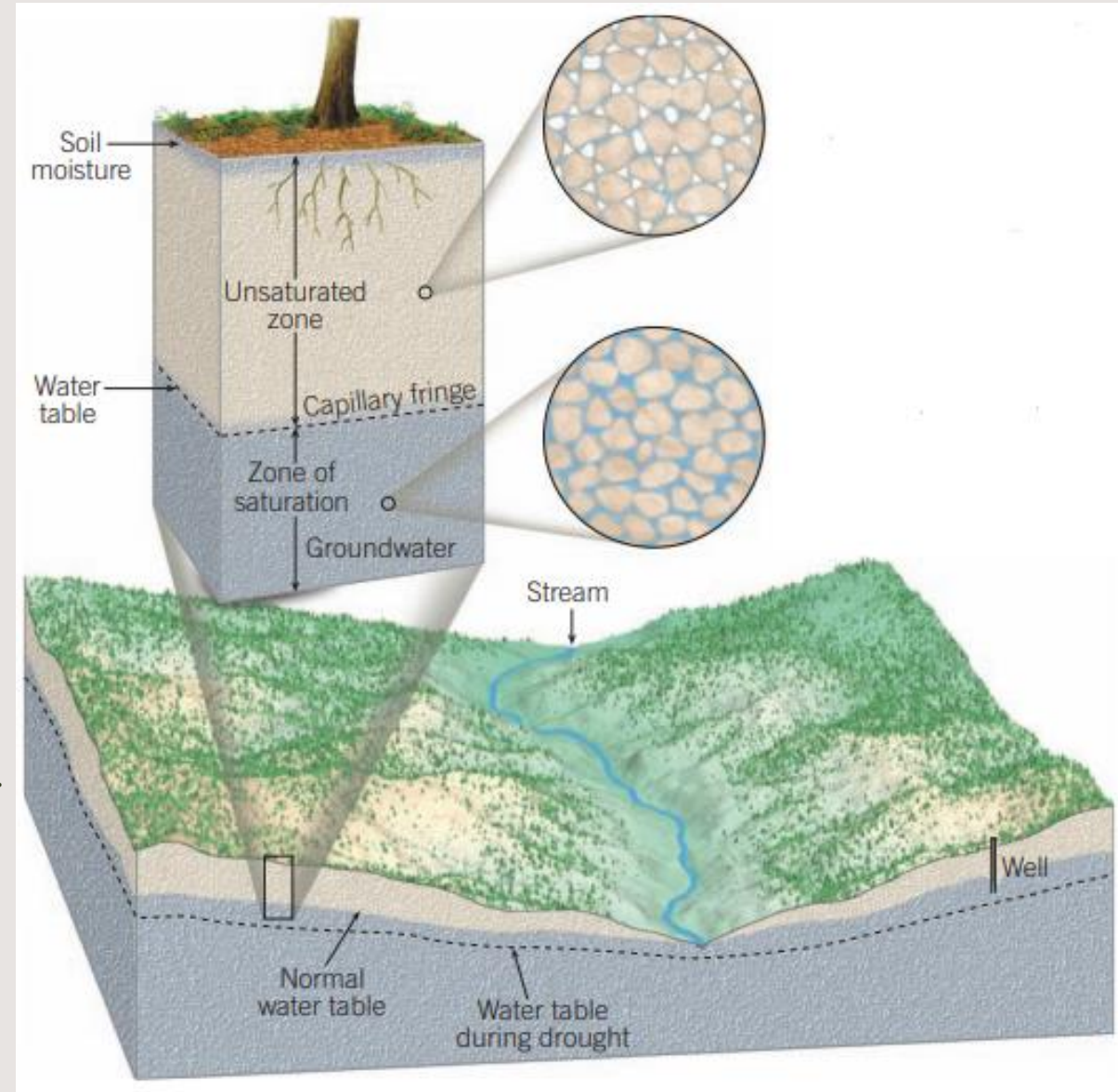
The Water Table

- It is the upper limit of the zone of saturation (Phreatic Zone).
- Its configuration depends from season to season, and on the amount of precipitation the area receives.
- Its subsurface shape is a subdued replica of the surface topography.

Figure 9.2 Water beneath Earth's surface

The shape of the water table is usually a subdued replica of the surface topography. During periods of drought, the water table falls, reducing streamflow and drying up some wells.

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Groundwater

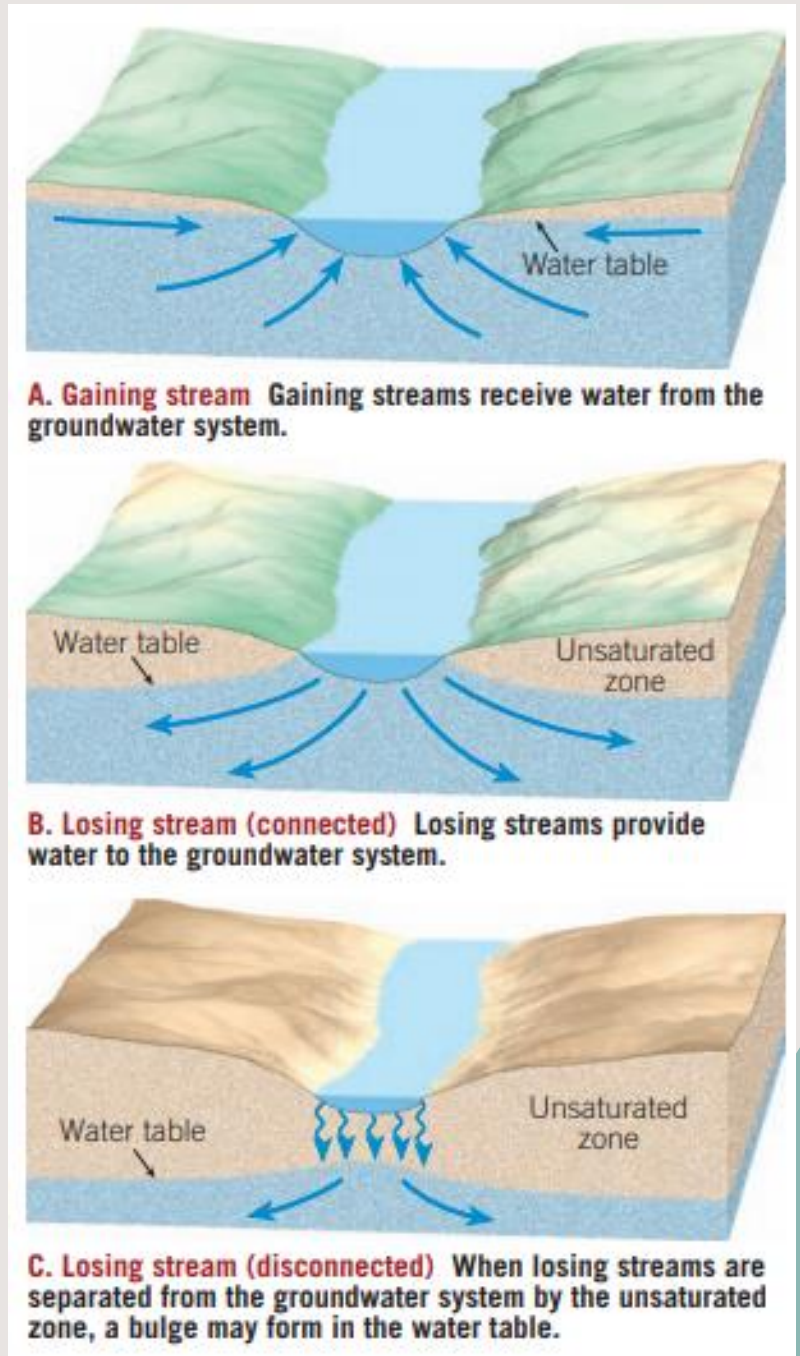
The Water Table

Ground Water and Stream Interaction

This interaction takes place in three possible ways:

1. **Gaining Stream** - for this to occur, the elevation of the water table must be higher than the level of the surface of the stream.
2. **Losing stream (connected)** - when this happens, the elevation of the water table must be lower than the surface of the stream. Connected to the groundwater system by a continuous saturated zone.
3. **Losing Stream (disconnected)** - they can be disconnected from the groundwater system by an unsaturated zone.

Figure 9.3 Interactions between the groundwater system and streams



Groundwater

The Water Table

Factors affecting ground water movement and storage:

1. **Porosity** - the ability of a rock of sediment to hold and retain water.
2. **Permeability** - refers to the ability to transmit fluids.

Aquitards - an impermeable layer that prevents water movement.

Aquifer - permeable rock strata or sediment that can freely transmit water.

TABLE 17.1 Selected Values of Porosity, Specific Yield, and Specific Retention*

Material	Porosity	Specific Yield	Specific Retention
Clay	50	2	48
Sand	25	22	3
Gravel	20	19	1
Limestone	20	18	2
Sandstone (semiconsolidated)	11	6	5
Granite	0.1	0.09	0.01
Basalt (fresh)	11	8	3

*Values in percent by volume.

Source: U.S. Geological Survey, Water Supply Paper 2220, 1987.

Groundwater

Groundwater Movement

Recharge zone/area - groundwater moving along flow paths from areas where groundwater is being replenished to a discharge area.

Discharge zone/area - a stream where groundwater is flowing back to the surface.

Drainage divide

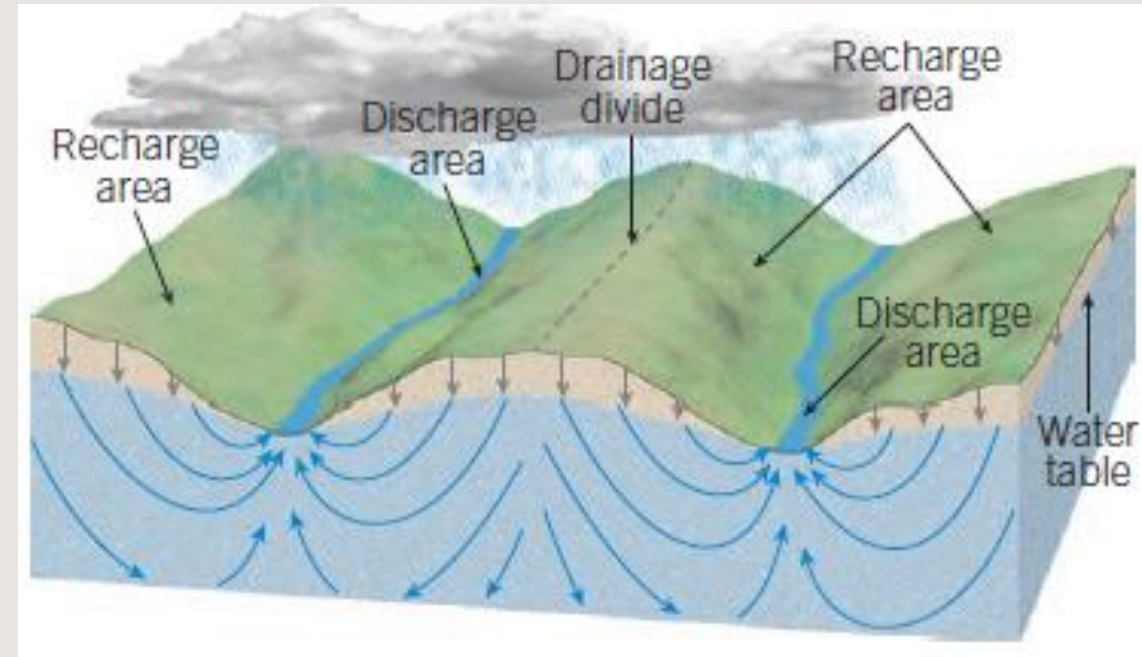


Figure 9.4 Groundwater movement

Arrows show paths of groundwater movement through uniformly permeable material.

Groundwater

Groundwater Movement Measurement

Darcy's equation

- First introduced by a French-scientist Engineer Henry Darcy
- Q is the discharge
- K is the permeability of the medium
- A is the area

$$Q = \frac{KA(h_1 - h_2)}{d}$$

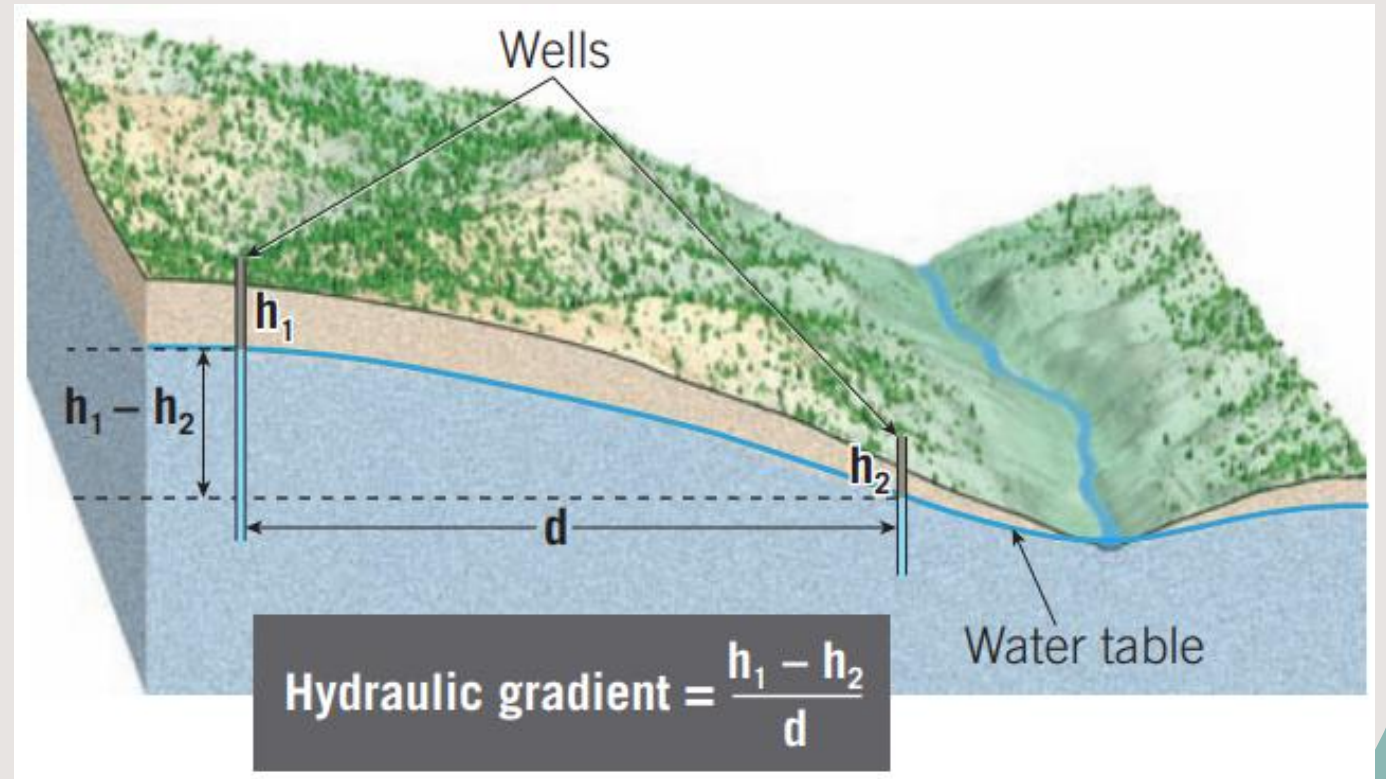


Figure 9.5 Hydraulic gradient

The hydraulic gradient is determined by measuring the difference in elevation between two points on the water table ($h_1 - h_2$) divided by the distance between them, d . Wells are used to determine the height of the water table.

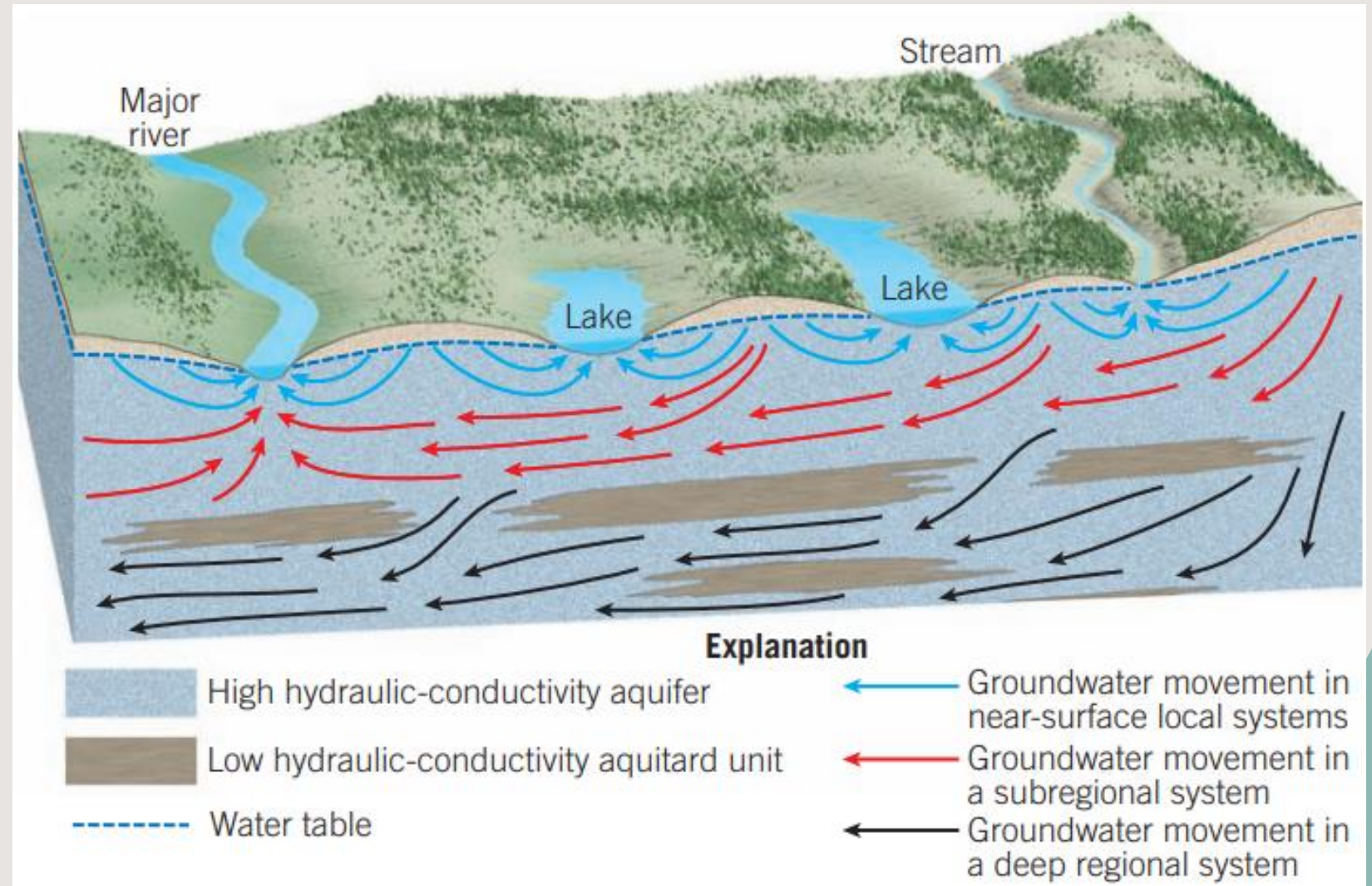
Groundwater

Groundwater Movement Measurement

Figure 9.6 Hypothetical groundwater flow system

The diagram includes subsystems at three different scales. Variations in Surface topography and subsurface geology can produce a complex situation. The horizontal scale of the figure could range from tens to hundreds of kilometers.

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Groundwater

Groundwater Extraction

Common method for groundwater extraction is through wells

Drawdown - term used to describe the lowering of the water table surrounding the well whenever water is being extracted from it.

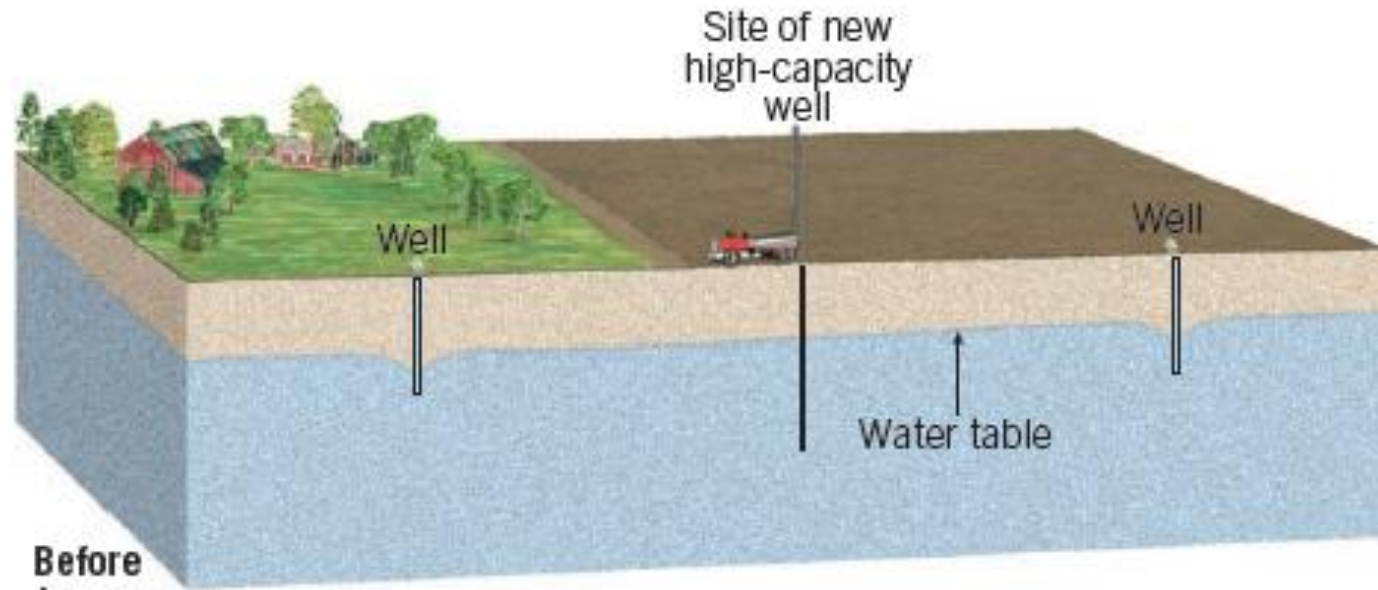
Cone of depression - a conical depression in the water table created by drawing the water out of the well. Because the cone of depression increases the hydraulic gradient near the well, groundwater will flow more rapidly toward the opening



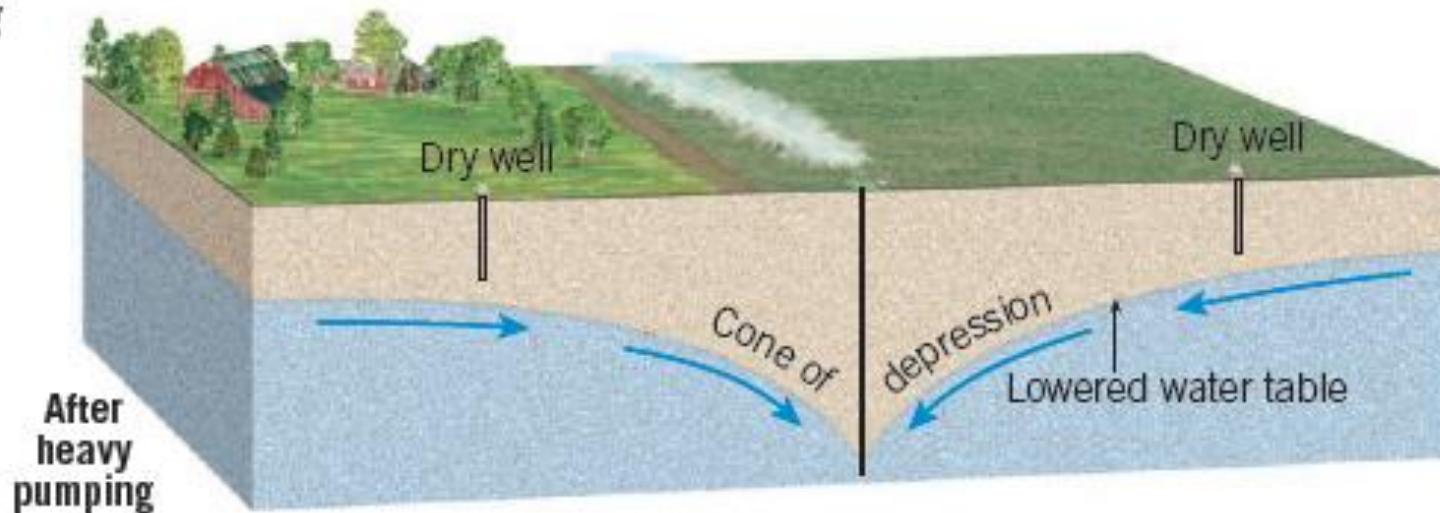
FIGURE 17.12 Wells

Wells are the most common means by which people obtain groundwater. (Top photo by Shutterstock; bottom photo by ASPYPPAGE Folzstock)





Before heavy pumping



After heavy pumping



SMARTFIGURE 17.13

Cone of depression

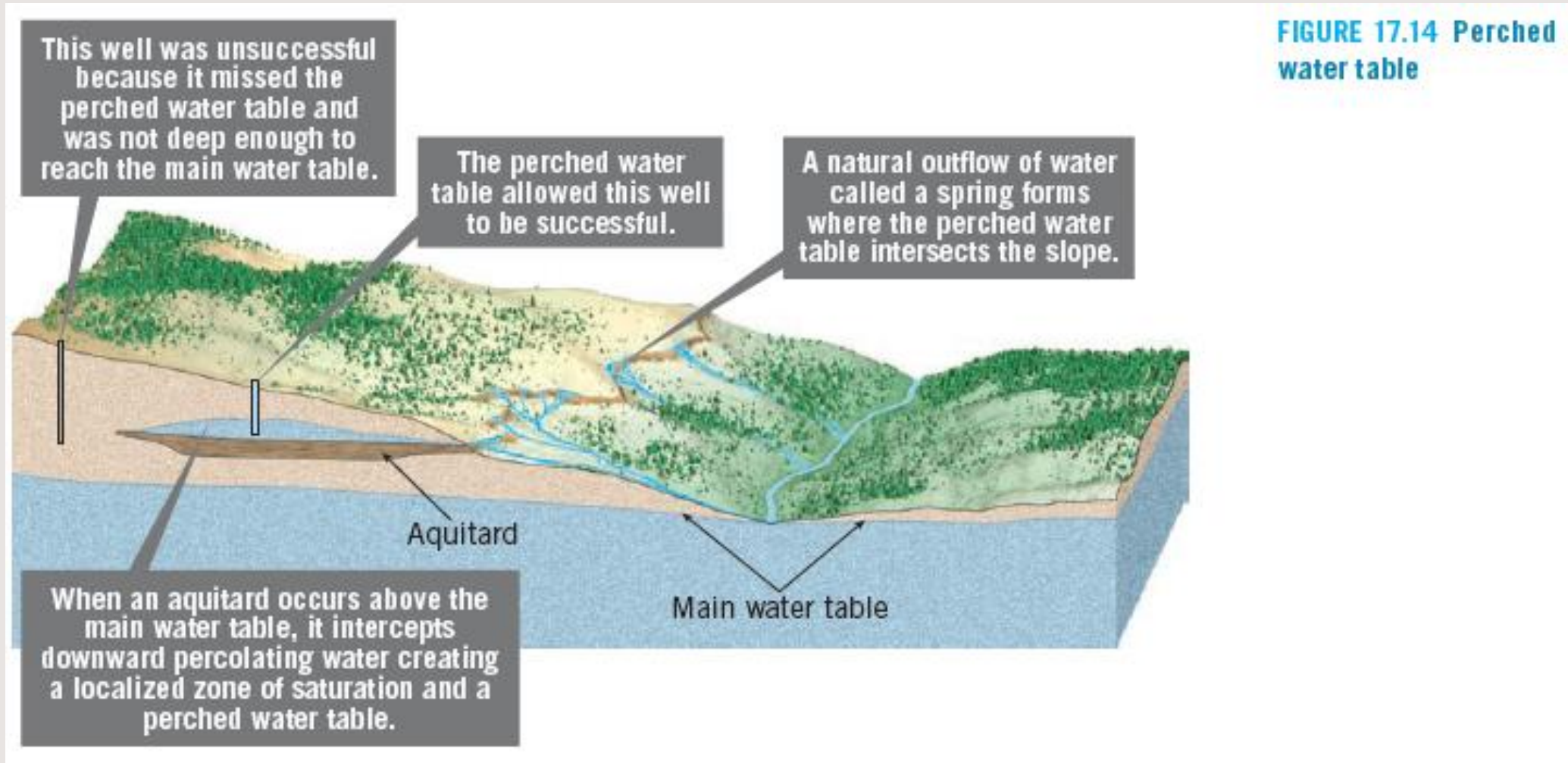
For most small domestic wells, the cone of depression is negligible. When wells are heavily pumped, the cone of depression can be large and may lower the water table such that nearby shallower wells may be left dry.



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Groundwater

Perched water table - forms where an aquitard is situated above the main water table.



Groundwater

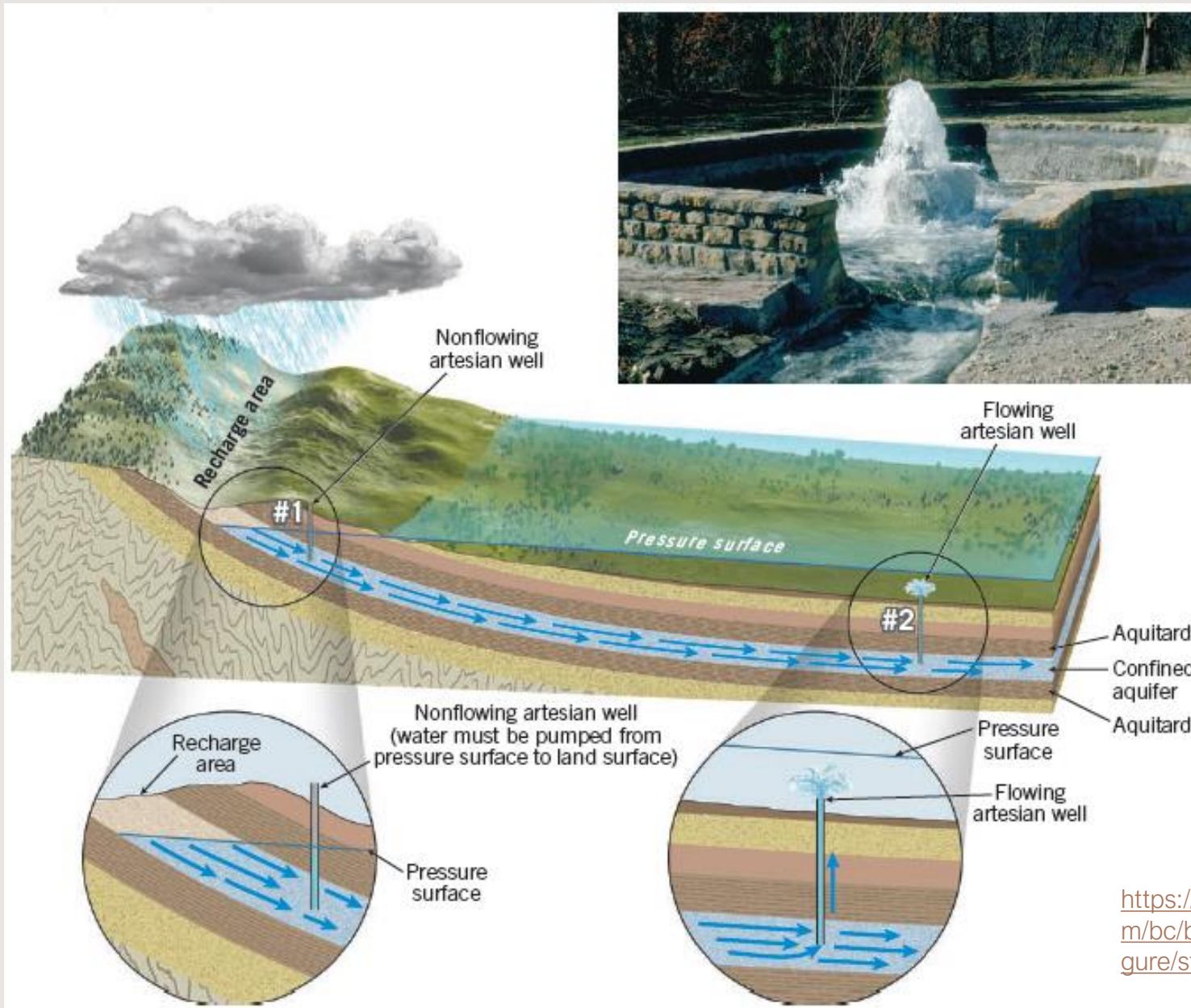
Artesian Systems - any situation in which groundwater under pressure rises above the level of the aquifer.

Types

Nonflowing artesian well - at this location, the pressure surface is below ground level

Flowing artesian well - when the pressure surface is above the ground and a well is drilled into the aquifer.

Groundwater



SMARTFIGURE 17.15 Artesian systems

These groundwater systems occur where an inclined aquifer is surrounded by impermeable beds (aquitards). Such aquifers are called *confined aquifers*. The photo shows a flowing artesian well.

(Photo by James E. Patterson)



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Groundwater

Springs, Hot Springs, Geysers

Springs – occurs when the water table intersects Earth's surface resulting a natural outflow of groundwater.

Hot Springs – when ground water becomes heated, and emerges to the surface as a natural outflow with a much higher temperature than the mean air temperature.

Geysers - are intermittent hot springs or fountains in which columns of water are ejected with great force at various intervals, often rising 30 to 60 meters (100 to 200 feet) into the air.

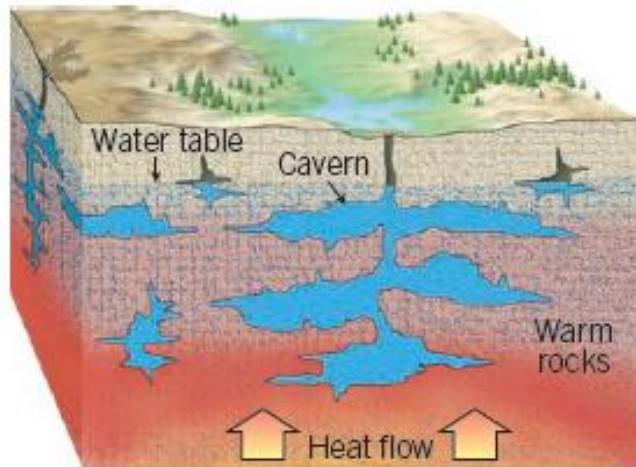


FIGURE 17.18 Thousand Springs
These well-known springs are along the Snake River in Hagerman Valley, Idaho. (Photo by David R. Frazier/Alamy Images)

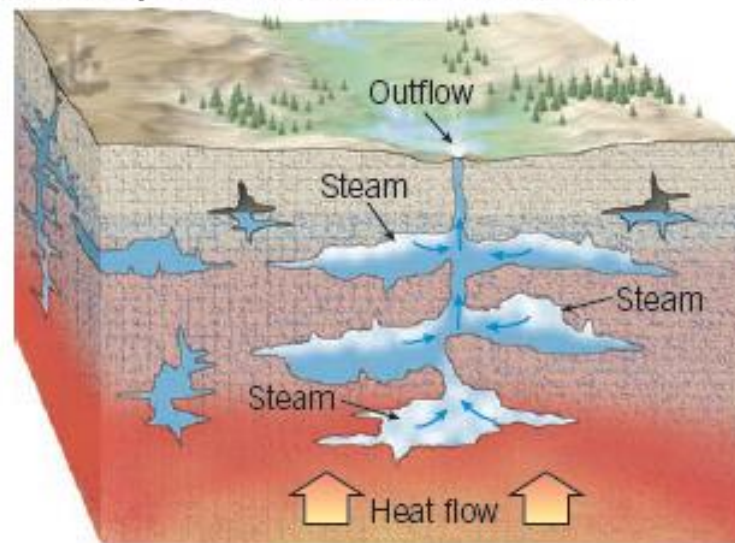


FIGURE 17.19 Hot springs
These examples represent the two different ways the water in hot springs is heated. (Top photo by New York Daily News/Getty Images; bottom photo by Andre Hasson/Alamy)

Water near the bottom is heated to near its boiling point. The boiling point is higher at the bottom because pressure is high due to the weight of all the water above.



Higher up in the geyser the water is also heated, therefore it expands causing some to flow out at the top. This outflow reduces the pressure on the water at the bottom.



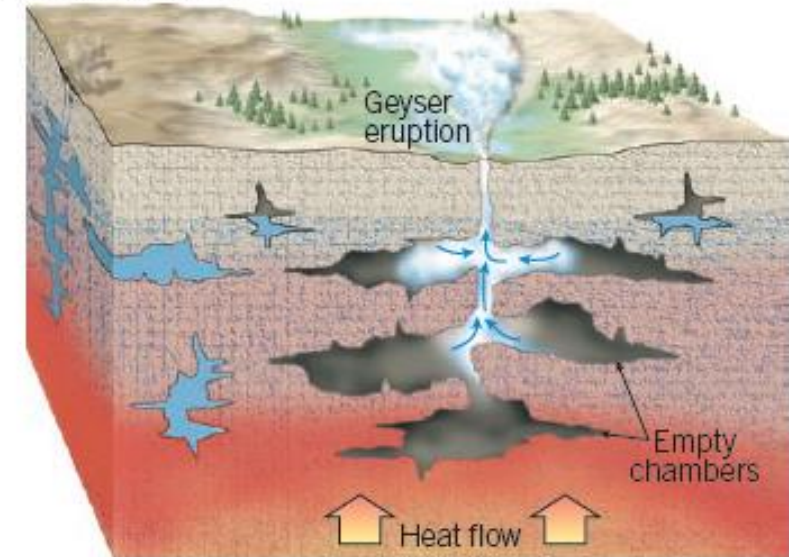
SMARTFIGURE 17.22 How a geyser works

A geyser can form if the underground plumbing does not allow heat to be readily distributed by convection.



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When pressure is reduced at the bottom, boiling occurs. Some of the bottom water flashes into steam. The expanding steam triggers an eruption.



Then the water flows back in and the whole process begins anew.

Groundwater

Environmental Problem - like many of our other valuable natural resources, groundwater is being exploited at an increasing rate. In some areas, overuse threatens the groundwater supply. In other places, groundwater withdrawal has caused the ground and everything resting on it to sink. Still other localities are concerned with the possible contamination of the groundwater supply.

Geologic Work of Groundwater - groundwater dissolves rock. This fact is key to understanding how caverns and sinkholes form. Because soluble rocks, especially limestone, underlie millions of square kilometers of Earth's surface, it is here that the groundwater carries on its important role as an erosional agent. Limestone is nearly insoluble in pure water but is quite easily dissolved by water containing small quantities of carbonic acid, and most groundwater contains this acid. It forms because rainwater readily dissolves carbon dioxide from the air and from decaying plants. Therefore, when groundwater comes in contact with limestone, the carbonic acid reacts with the calcite (calcium carbonate) in the rocks to form calcium bicarbonate, a soluble material that is then carried away in solution.

Reference

- Tarbuck, Lutgens and Tasa (2014), Earth: An Introduction to Physical Geology 11th ed., Pearson Education, Inc